The algebraic fraction. Cancelling fractions

A fraction whose parts are polynomials — and the one rule that only factors may be cancelled.

LESSON 112–113 | 2 × 40 MIN | MATEMATIKA 7, §50 | S8 7

LESSON CLOCK

15' | 20' | 25' | 20' | 5'

What an algebraic fraction is	15 min
Where it has no value	20 min
Cancelling	25 min
Factors, never terms	20 min
Homework	5 min

LEARNING OBJECTIVES

- Define an algebraic fraction and state where it has no value.
- Factorise numerator and denominator before cancelling.
- Cancel only common factors, never terms.
- Write the values of the letter that must be excluded.

TEXTBOOK

National	pp. 327–334
Cambridge	Stage 8, pp. 66–78
Workbook	Exercise 7.1

01 Explanation

What an algebraic fraction is

An **algebraic fraction** is a quotient of two polynomials, exactly as an ordinary fraction is a quotient of two whole numbers.

$$\frac{P(x)}{Q(x)} \text{ with } Q(x) \neq 0$$

EXPRESSION	AN ALGEBRAIC FRACTION?
$\frac{x+1}{x-2}$	yes
$\frac{x^2}{3}$	yes — the denominator may be a number
$\frac{5}{x}$	yes
$x^2 + 1$	a polynomial, not a fraction

EVERYTHING YOU KNOW ABOUT ORDINARY FRACTIONS CARRIES OVER

Cancelling, common denominators, adding, multiplying, dividing — the rules are the same ones learnt in Quarter I. Only the objects in the numerator and denominator have changed.

Where it has no value

A denominator of zero makes the fraction meaningless, so those values of the letter must be excluded and written down.

FRACTION	DENOMINATOR ZERO WHEN	EXCLUDED
$\frac{5}{x}$	$x = 0$	$x \neq 0$
$\frac{x + 1}{x - 2}$	$x = 2$	$x \neq 2$
$\frac{3}{x^2 - 9}$	$x = \pm 3$	$x \neq 3, x \neq -3$
$\frac{x}{x^2 + 1}$	never	nothing to exclude

FIND THE EXCLUDED VALUES *BEFORE* CANCELLING

In $\frac{x^2 - 9}{x - 3}$ the answer is $x + 3$, but $x = 3$ is still forbidden — the original fraction had no value there. Cancelling does not make a forbidden value legal.

Cancelling

Factorise both parts fully, then divide out every factor they share.

FRACTION	FACTORISED	SIMPLIFIED
$\dfrac{12a^3}{4a}$	$\dfrac{4a \cdot 3a^2}{4a}$	$3a^2$
$\dfrac{x^2 - 9}{x + 3}$	$\dfrac{(x - 3)(x + 3)}{x + 3}$	$x - 3$
$\dfrac{x^2 + 5x + 6}{x + 2}$	$\dfrac{(x + 2)(x + 3)}{x + 2}$	$x + 3$
$\dfrac{2x + 6}{x^2 - 9}$	$\dfrac{2(x + 3)}{(x - 3)(x + 3)}$	$\dfrac{2}{x - 3}$

THIS IS WHY THE LAST CHAPTER CAME FIRST

Every one of these cancellations begins with a factorisation from the abridged formulae. Without that chapter, none of these fractions could be simplified at all.

Factors, never terms

ATTEMPT	RIGHT?	WHY
$\dfrac{(x - 3)(x + 3)}{x + 3} = x - 3$	yes	$x + 3$
$\dfrac{x^2 - 9}{x + 3} = x^2 - 3$	no	the 9 and the 3 are not factors
$\dfrac{x + 5}{x} = 5$	no	x
$\dfrac{2x}{2} = x$	yes	2

A TERM THAT IS ADDED CAN NEVER BE CANCELLED

$\dfrac{x + 5}{x}$ is not 5: try $x = 1$ and the fraction is 6. Only things that are **multiplied** may be divided out.

02 Worked examples

EXAMPLE 1 **Simplify $\dfrac{x^2 + 5x + 6}{x + 2}$ and state the excluded value.**

1. $x + 2 = 0$ when $x = -2$, so $x \neq -2$.

   Written down first.

2. $x^2 + 5x + 6 = (x + 2)(x + 3)$

   Product 6, sum 5.

3. $\dfrac{(x + 2)(x + 3)}{x + 2}$

4. $= x + 3$

ANSWER

$x + 3,\ x \neq -2$

EXAMPLE 2 **Simplify $\dfrac{2x + 6}{x^2 - 9}$.**

1. $2x + 6 = 2(x + 3)$

2. $x^2 - 9 = (x - 3)(x + 3)$

3. Cancel $x + 3$.

4. $= \dfrac{2}{x - 3}$, with $x \neq \pm 3$.

ANSWER

$\dfrac{2}{x - 3}$

EXAMPLE 3 **Is $\frac{x+5}{x} = 5$ correct?**

1. Try $x = 1$: the fraction is 6.
2. So the equality is false.
3. x is a term of the numerator, not a factor of it.

ANSWER

No

03 Interactive model

Write (x + 5)/x = 5 on the board and ask the class to test it with x = 1; one substitution ends the argument for the rest of the year.

Factorise, then cancel

$$\frac{x^2 + 5x + 6}{x + 2}$$

1. $x \neq -2$

The excluded value, noted first.

2. $x^2 + 5x + 6 = (x + 2)(x + 3)$

Product 6, sum 5.

3. $x + 3$

Cancel the common factor $x + 2$.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Algebraic fraction	Algebraik kasr	Алгебраическая дробь
Numerator	Surat	Числитель
Denominator	Maxraj	Знаменатель
To cancel	Qisqartirish	Сокращать
Common factor	Umumiy ko‘paytuvchi	Общий множитель
Excluded value	Mumkin bo‘lmagan qiymat	Недопустимое значение
Undefined	Aniqlanmagan	Не определено
Simplest form	Eng sodda ko‘rinish	Несократимый вид

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

An algebraic fraction is a quotient of:

- two numbers
- two polynomials
- two equations
- two roots

$\frac{5}{x}$ has no value when:

- $x = 5$
- $x = 0$
- $x = 1$
- never

$\frac{3}{x^2 - 9}$ excludes:

- $x = 3$
- $x = -3$
- both
- neither

$\frac{x^2 - 9}{x + 3}$ simplifies to:

- $x^2 - 3$
- $x - 3$
- $x + 3$
- x

$\frac{x + 5}{x}$ simplifies to:

- 5
- $x + 5$
- it does not simplify
- 6

Cancelling is allowed for:

- terms
- factors
- anything
- signs

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\dfrac{12a^3}{4a}$

ANSWER $3a^2$

2. $\dfrac{x^2 - 9}{x + 3}$

ANSWER $x - 3$

3. $\dfrac{2x}{2}$

ANSWER x

4. $\dfrac{6x^2}{3x}$

ANSWER $2x$

5. $\dfrac{5}{x}$: the excluded value

ANSWER $x = 0$

6. $\dfrac{x + 1}{x - 2}$: the excluded value

ANSWER $x = 2$

7. $\dfrac{x^2 + 5x + 6}{x + 2}$

ANSWER $x + 3$

Medium · the standard question

7 PROBLEMS

1. $\dfrac{2x + 6}{x^2 - 9}$

ANSWER $\dfrac{2}{x - 3}$

2. $\dfrac{x^2 - 4}{x^2 + 4x + 4}$

ANSWER $\dfrac{x - 2}{x + 2}$

3. $\dfrac{3}{x^2 - 9}$: the excluded values

ANSWER $x = \pm 3$

4. $\dfrac{x^2 - 7x + 12}{x - 3}$

ANSWER $x - 4$

5. $\dfrac{4x^2 - 25}{2x + 5}$

ANSWER $2x - 5$

6. $\dfrac{x}{x^2 + 1}$: the excluded values

ANSWER None

7. Is $\dfrac{x + 5}{x} = 5$?

ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. $$\frac{x^3 - 8}{x - 2}$$

ANSWER $x^2 + 2x + 4$

2. $$\frac{x^2 - 1}{x^2 + 2x + 1}$$

ANSWER $\frac{x - 1}{x + 1}$

3. $$\frac{2x^2 - 18}{x^2 + 6x + 9}$$

ANSWER $\frac{2(x - 3)}{x + 3}$

4. $$\frac{a^2 - b^2}{a^2 - 2ab + b^2}$$

ANSWER $\frac{a + b}{a - b}$

5. $$\frac{x^2 + x - 6}{x^2 - 4}$$

ANSWER $\frac{x + 3}{x + 2}$

6. Why must the excluded values be found before cancelling?

ANSWER The original fraction, not the simplified one, defines them

7. $$\frac{3 - x}{x - 3}$$

ANSWER -1

07 Homework

Homework — 5 tasks

Write the excluded values first, then simplify.

1. Simplify $\dfrac{x^2 - 16}{x + 4}$ and state the excluded value.
2. Simplify $\dfrac{3x + 12}{x^2 - 16}$.
3. Simplify $\dfrac{x^2 + 7x + 10}{x + 5}$.
4. Find the excluded values of $\dfrac{7}{x^2 - 4x}$.
5. Explain with one substitution why $\dfrac{x + 3}{x}$ is not 3.